

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$$\frac{1}{x^2+10x+25} = 4$$

If x is a solution to the given equation, which of the following is a possible value of $x+5$?

Answer

- A. $\frac{1}{2}$
- B. $\frac{5}{2}$
- C. $\frac{9}{2}$
- D. $\frac{11}{2}$

Correct Answer: A

Rationale

Choice A is correct. The given equation can be rewritten as $\frac{1}{(x+5)^2} = 4$. Multiplying both sides of this equation by $(x+5)^2$ yields $1 = 4(x+5)^2$. Dividing both sides of this equation by 4 yields $\frac{1}{4} = (x+5)^2$. Taking the square root of both sides of this equation yields $\frac{1}{2} = x+5$ or $-\frac{1}{2} = x+5$. Therefore, a possible value of $x+5$ is $\frac{1}{2}$.

Choices B, C, and D are incorrect and may result from computational or conceptual errors.